

**TELECONNECTION BETWEEN ANTARCTIC
SEA ICE EXTENT AND PRECIPITATION IN
PENINSULAR MALAYSIA**

TAN QIAO ROU

UNIVERSITI SAINS MALAYSIA

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TELECONNECTION BETWEEN ANTARCTIC SEA ICE EXTENT AND PRECIPITATION IN PENINSULAR MALAYSIA

by

TAN QIAO ROU

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DECLARATION BY AUTHOR

This dissertation is composed of my original work and contains no material previously published or written by another person except where due reference has been made in the text. The content of my dissertation is the result of work I have carried out since the commencement of my research project and does not include a substantial part of work that has been submitted to qualify for the award of any other degree or diploma in any university or other tertiary institution.



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LIST OF ABBREVIATION

ACW	Antarctic Circumpolar Wave
ENSO	El Niño–Southern Oscillation
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA5	Fifth-generation ECMWF Atmospheric Reanalysis
ITCZ	Intertropical Convergence Zone
MJO	Madden–Julian Oscillation
NSIDC	National Snow and Ice Data Center
OLR	Outgoing Longwave Radiation
PV	Potential Vorticity
QG	Quasi-Geostrophic
SIA	Sea Ice Area
SIC	Sea Ice Concentration
SIE	Sea Ice Extent
SST	Sea Surface Temperature
SAM	Southern Annular Mode
SO	Southern Ocean

List of Symbol

v	Meridional Wind Component
ω	Vertical Velocity
u	Zonal Wind Component

TELEKONEKSI ANTARA KELUASAN AIS LAUT ANTARTIKA DAN VARIASI HUJAN DI SEMENANJUNG MALAYSIA

ABSTRAK

Ais laut Antartika memainkan peranan penting dalam sistem iklim global dan berupaya mempengaruhi corak cuaca di kawasan yang jauh dari kawasan kutub melalui proses telekoneksi atmosfera. Walaupun kajian terdahulu menunjukkan bahawa keluasan ais laut Antartika (Sea Ice Extent, SIE) memberi kesan terhadap taburan hujan di kawasan seperti India dan Asia Timur, pengaruhnya terhadap variasi hujan di Malaysia masih belum dikaji secara meluas. Oleh itu, kajian ini bertujuan untuk menyiasat hubungan antara SIE Antartika di sektor Lautan Hindi dengan variasi hujan bermusim di Semenanjung Malaysia. Analisis korelasi Pearson digunakan untuk mengenal pasti hubungan yang signifikan antara SIE dan hujan, manakala analisis perbezaan komposit digunakan untuk mengkaji perubahan dalam peredaran atmosfera yang berkaitan. Beberapa pembolehubah atmosfera utama, termasuk ketinggian geopotensi, angin mendatar, halaju menegak, dan perbezaan kelembapan bersepadu secara menegak, dianalisis bagi menerangkan mekanisme fizikal yang menghubungkan anomali ais laut Antartika dengan hujan di kawasan tropika. Hasil kajian menunjukkan bahawa hubungan antara SIE dan hujan adalah bergantung kepada musim, dengan bulan Februari menunjukkan hubungan yang paling kuat dan konsisten secara spatial. Peningkatan SIE Antartika didapati berkait dengan pengurangan hujan di Semenanjung Malaysia, yang disebabkan oleh pelarasan peredaran atmosfera berskala besar seperti aktiviti gelombang Rossby pegun, penumpuan di lapisan atas atmosfera, pergerakan udara menurun (subsiden), dan pengurangan penumpuan kelembapan. Sebaliknya, hubungan pada bulan Julai dan Ogos adalah lebih lemah dan tidak konsisten kerana pengaruh proses monsun serantau yang lebih dominan. Secara

keseluruhan, kajian ini menunjukkan bahawa telekoneksi antara Antartika dan kawasan tropika adalah bergantung kepada keadaan atmosfera dan lebih ketara apabila atmosfera tropika berada dalam keadaan sensitif secara dinamik. Dapatan kajian ini menekankan kepentingan mempertimbangkan perubahan ais laut di latitud tinggi dalam memahami dan mentafsir variasi corak hujan di Semenanjung Malaysia.

TELECONNECTION BETWEEN ANTARCTIC SEA ICE EXTENT AND PRECIPITATION IN MALAYSIA

ABSTRACT

Antarctic sea ice plays an important role in the global climate system and can influence weather patterns beyond the polar regions through atmospheric teleconnections. Although previous studies have shown that Antarctic sea ice extent (SIE) affects rainfall in regions such as India and East Asia, its influence on rainfall variability in Malaysia remains less understood. This study investigates the relationship between Antarctic SIE in the Indian Ocean sector and seasonal rainfall variability over Peninsular Malaysia. Pearson correlation analysis was applied to identify significant relationships between SIE and rainfall, while composite difference analysis was used to examine associated atmospheric circulation patterns. Key atmospheric variables, including geopotential height, horizontal winds, vertical velocity, and vertically integrated moisture divergence, were analysed to explain the physical mechanisms linking Antarctic sea ice anomalies to tropical rainfall. The results show that the SIE–rainfall relationship is strongly during February exhibits the most robust and spatially coherent signal. This negative correlation between the precipitation in Peninsular Malaysia is linked to large-scale circulation adjustments, including stationary Rossby wave activity, upper-level convergence, tropospheric subsidence, and reduced moisture convergence. In contrast, the relationship during July and August is weaker and less consistent, as regional monsoon processes dominate rainfall variability during these months. Overall, this study demonstrates that Antarctic–tropical teleconnections are regime dependent and are most evident when the tropical atmosphere is dynamically sensitive. These findings highlight the importance of considering high-latitude cryospheric variability when interpreting rainfall pattern in Peninsular Malaysia.

CHAPTER 1 INTRODUCTION

1.1 Research Background

Sea ice is one of the compositions in polar region, its significant and bring uncertainty implications to remote the global climate system. Based on the (Xiao et al., 2024) research paper mentioned that there are many researchers conducted the research to show the connection between air-ocean parameters and indices. For example, Southern Annular Mode (SAM), El Niño and Southern Oscillation (ENSO), Southern Indian Ocean Dipole and Atlantic Meridional Mode and tropical precipitation. According to the paper (Azhar et al., 2023a), Antarctic sea ice extent (SIE) is a sensitive climate variable that influences the global climate system across both hemispheres through various atmospheric-oceanic teleconnections. It works by exciting meridional wave trains along the Indian Ocean that alter the strength of the Polar, Ferrel, and Hadley cells, which in turn modulates convective activity and precipitation over regions like India.

This paper (Goosse et al., 2018) highlights that climate dynamics in polar regions are governed by complex radiative and non-radiative interactions between the atmosphere, ocean, sea ice, ice sheets, and land surfaces. It emphasizes that quantifying these multi-component interactions is essential for understanding the processes responsible for polar climate change and reducing uncertainties in model projections. Antarctic circumpolar wave (ACW) was first introduced by the White and Peterson (1996) as a system of coupled oceanic and atmospheric anomalies circling the Southern Ocean which shown the significant teleconnection with global climate pattern. It can be shown by the Meridional Wind Stress and Net Heat Flux drive the subsequent anomalies in Sea-Ice Extent (SIE) and Sea Surface Temperature (SST). There are two main waves which can be described as in phase and out of phase, in

phase represent the northward sea-ice extent while out of phase is the southward sea-ice retreat (Y. Zhang & Semtner, 2001).

The climate oscillation is related to the equatorial wave dynamic which means wave signal traverse the equator that establishing the connection between Southern and Northern Hemisphere. The flux vector tilt of a zonal wave number less than 6 will lead to its dissipation or absorption at a critical layer near the equator (Xiao et al., 2024). A barotropic stationary Rossby wave is crucial to carry the wave energy to across the equator which guide the weather system. Based on (Y. Li et al., 2019a), the Rossby wave can either propagate or blocked by reflection or absorption at near the equatorial zone to transfer energy from one hemisphere to another hemisphere. The anomalies in one hemisphere creates a contrast weather and climate system in the other. The author said that the anomalies of climate and weather respond are vice versa between Southern Hemisphere and Northern Hemisphere.

El Niño-Southern Oscillation (ENSO) are one of the climate patterns that can correlate with Malaysia dry and wet season. Malaysia extreme rainfall event is affected by El Niño and La Niña but not a linear correlation. In this paper (Tangang et al., 2017a), author studied the relationship between ENSO and extreme precipitation events because the impact of ENSO on mean precipitation in different region of world has been studied but relate to extreme precipitation events is rare. For Maritim Continent there is high (low) possibility for extreme rainfall happen during La Niña (El Niño). The ENSO cycle happen from July to September but this study showed the ENSO reach its mature phase from October to December as mentioned in introduction of the paper (Tangang et al., 2017b). Furthermore, the mechanism of extreme rainfall over Peninsular Malaysia and west Borneo is from November to December and January to February due to the strong tropical easterlies that propagating from

Philippine area and Borneo. The propagation of the wind system during northeast monsoon is fundamentally a response to the seasonal shift of the Near-Equator Trough (NET).

Malaysia rainy period happen during northeast monsoon (December to February) and dry period occur during southwest monsoon (June to August). Normally, precipitation distribution over Malaysia is influenced by monsoon cycles and local topography. The shift in precipitation distribution is controlled by the altering position or intensification of the Intertropical Convergence Zone (ITCZ) which also influence the monsoon system and hydrological system (Prabhu et al., 2010). Based on the paper of (Mamalakis et al., n.d.), it is important to understand ICTZ mechanism that response to the large-scale climate variability and anthropogenic forcing which aid in identify the precipitation pattern. Its position is also controlled by tropical mechanisms that impact sea surface temperature (SST) gradients. The hemisphere energy asymmetries is the key factor to the shift the position of ITCZ which tend to move to warmer hemisphere. Normally, scientises studied global average (zonal-mean), the ITCZ actually shifts differently depending on the longitude. Under climate change scenarios, it is projected to shift northward over the Indian Ocean and eastern Africa, while shifting southward in the eastern Pacific and Atlantic Oceans.

Besides, the Southern Annular Mode (SAM) is known as an atmospheric circulation pattern in Southern Hemisphere that significantly control the strength of westerlies and it also known as Antarctic Oscillation (AAO) (Lefebvre et al., 2004). Hence, it acts as a bridge to link the high latitude Antarctic atmospheric circulation to the climate variability in Indian Ocean region. During positive SAM phase since the early 1900s, the land and sea thermal contrast as well as the anomalous vertical motion is promoted to intensify the southerly cross-equatorial flow.(Huang et al., 2021). The

Southern Annular Mode (SAM) is characterized by synchronous pressure anomalies of opposite signs between Antarctica and the mid-latitudes.

Madden-Julian Oscillation (MJO) is one of the dominant component of intraseasonal variability in the tropical atmosphere and its cycle is lasting to 30-90 days. It is a planetary-scale system involving a large-scale coupling between atmospheric circulation and deep convection. While the convective center propagates eastward at an average speed of approximately 5 m/s across the Indian and Pacific Oceans, the entire life cycle of a single MJO event can last over a month (Zhang, 2005).

Based on the research published by (J. Li et al., 2024) The identification and forecasting of regional heavy rainfall depend heavily on two primary physical factors: moisture supply and vertical motion. While moisture provides the necessary fuel for precipitation, vertical motion serves as the dynamic mechanism that triggers and locates the rainfall. Understanding these teleconnections requires a detailed look at how remote drivers alter local moisture transport pathways and vertical velocity, similar to the process-based analysis used to distinguish extreme rainfall types in midlatitude regions.

Whether rainfall increases or decreases depends on moisture availability, depth of ascent, and dynamical context. Regions of ascent and descent are associated with divergence and convergence of the horizontal wind, not simply with lows and highs in the pressure field. The occurrence of rainfall is determined by the low level convergence, high level divergence and deep convection coherence. However, the high intensity of rainfall depends on the rising motion and the moisture flux. Vertical motion is forced primarily by horizontal divergence and convergence, rather than directly by the pressure field itself. Vertical motion in the atmosphere is dynamically forced by horizontal divergence and convergence associated with vorticity and thermal advection,

rather than directly by pressure or geopotential height anomalies (Holton & Hakim, 2013). The deeply rising motion is caused by the upper level divergence but it can be blocked by the upper level divergence ends up the sinking air formed. The vertical motion will control the cloud development which act as the activator for precipitation.

Atmospheric motions can be classified into planetary, synoptic, mesoscale, and microscale regimes based on their characteristic spatial and temporal scales, each associated with distinct dynamical balances. At the planetary scale, large-scale circulation patterns such as jet streams and Rossby waves regulate the background flow of the atmosphere, while synoptic-scale systems, including troughs and ridges, are primarily responsible for generating regions of ascent and subsidence through horizontal wind convergence and divergence. Mesoscale and microscale processes further modulate local weather but are often embedded within larger-scale circulation patterns. (John M. Wallace & Peter V. Hobbs, 2006).

The interaction between the atmospheric circulations of the Northern and Southern Hemispheres is a critical component of global climate teleconnections. This interhemispheric coupling is largely driven by stationary Rossby waves, which transport wave energy across the equator. However, the equatorial region often acts as a "critical line" or barrier, reflecting or absorbing these waves rather than allowing them to pass through (Y. Li et al., 2019b).

Despite the established understanding of individual climate drivers such as ENSO and the SAM, the specific teleconnection between Antarctic sea ice extent (SIE) within the Indian Ocean sector and the precipitation patterns over Malaysia remains complex and less explored. While previous research highlights the role of Rossby wave propagation and interhemispheric coupling, the precise atmospheric mechanisms showed SIE anomalies modulate vertical motion and moisture transport over the

Maritime Continent which require further quantification. Therefore, this study aims to investigate the correlation between Indian Ocean sector sea ice and Malaysian rainfall variability. By analysing the dynamical interactions between high-latitude forcing and tropical convective activity, this research seeks to clarify how polar climate signals traverse the equatorial barrier to influence regional hydrological cycles, ultimately contributing to more accurate precipitation forecasting in Malaysia.

1.2 Problem Statement

Antarctica may be thousands of miles away, but its sea ice acts like a giant "control center" that helps regulate how air and water move across the entire planet. Scientists have already studied how changes in this ice affect the weather in big countries like China and India (Azhar et al., 2023b; Makula & Zhou, 2023; Prabhu et al., 2009, 2021). At present, there is insufficient research examining the relationship between the extent of Antarctic sea ice and precipitation trends in Malaysia. As climate change affects Antarctic sea ice patterns, these changes may have significant implications for extreme rainfall events and contribute to rising sea levels in the global climate system. The absence of research on the teleconnection between Antarctic sea ice extent and precipitation in Malaysia reveals a significant gap in our understanding of how global climate changes can affect local weather patterns. Addressing this gap is crucial for enhancing climate forecasting, flood preparedness, and the management of water resources in the area.

1.3 Objectives

- To investigate the influence of Antarctic sea ice extent anomalies on seasonal rainfall variability in Malaysia through large-scale atmospheric circulation patterns.

- To determine the spatial and temporal correlation between Antarctic sea ice extent anomalies and monthly rainfall variability in Peninsular Malaysia.
- To identify the teleconnection which link the Antarctic sea ice extent anomaly and the monthly rainfall pattern in Peninsular Malaysia via large-scale atmospheric circulation.

1.4 Scope of Study

1. The Antarctic Sea Ice region is large and National Snow and Ice Data Center (NSIDC) split the sea ice data into 5 regions and the region near to Malaysia is Indian Ocean sector.
2. The atmospheric parameters are used as to discuss the teleconnection geopotential height, zonal wind, meridional wind, vertically integrated moisture divergence, vertical velocity and

CHAPTER 2 LITERATURE REVIEW

Based on the geography, Malaysia is situated between two large ocean which are Pacific Ocean and Indian Ocean so Malaysia climate is link to the natural climate phenomena. Sea ice extent bring impact to the atmospheric and oceanic parameter and sea ice condition is characterized by the Antarctic Circumpolar wave (ACW). ACW is linked to large -scale atmospheric circulation so it determines the global climate such as Indian summer monsoon and the quantum of the monsoon that linked to the severity of drought or flood(Prabhu et al., 2010).

The research conducted by (Chandra & Sandeep, 2024) found that the Indian summer monsoon is highly sensitive to the condition change of Antarctic sea ice compared to the Arctic. The respond of India summer monsoon to the polar sea ice melt result in decreasing the intensity precipitation due to the weak circulation. The Southwest Monsoon refers to the moisture-laden winds blowing from the southwest Indian Ocean that bring the rainy season (B. Wang et al., 2009).

The MJO is driven by complex interactions between moisture-convection feedback and dynamical theories such as Wind-Induced Surface Heat Exchange (WISHE). According to (C. Zhang, 2005), WISHE suggests that surface evaporation is enhanced to the east of the convective center by mean easterly winds, while low-level moisture convergence provides the necessary fuel for deep convection. However, this eastward propagation often encounters a "barrier" when reaching the Maritime Continent (MC). Zhang (2005) noticed that the MJO signal typically weakens in this region due to topographic interference and competition from the strong diurnal cycle of local thunderstorms, which can deplete the moisture and energy required by the larger-scale MJO system.

According to Robertson et al. (2011), the complex topography of the Maritime Continent triggers critical wind-terrain interactions that drive weather and climate variability across multiple scales. These regional-scale processes modulate the impact of the El Niño-Southern Oscillation (ENSO). While El Niño events are typically associated with a significant delay in monsoon onset and reduced seasonal rainfall, research reveals that rainfall intensity over orography can actually increase. This is attributed to strengthened diurnal land-sea and mountain-valley breezes occurring under the weakened large-scale wind conditions characteristic of El Niño.

Recent studies indicate that the debated ENSO–NAO relationship arises from strong subseasonal variability within boreal winter. Zhang and Jiang (2023) showed that ENSO is positively associated with the NAO in early winter but reverses in late winter due to delayed warming of the North Tropical Atlantic and associated subtropical convection. Apart from that, Qiao et al. (2020) found that circulation prediction skill and ENSO–NAO coherence are stronger in February than in January, highlighting late winter as a period of enhanced teleconnection robustness.

The Southern Annular Mode (SAM) serves as the primary driver of atmospheric variability in the Southern Hemisphere, where its positive phase is defined by relatively low pressure over Antarctica compared to the mid-latitudes. This shift strengthens the meridional pressure gradient, intensifying the circumpolar westerlies and altering high-latitude circulation patterns that can influence global teleconnections, including regional sea ice distribution and tropical moisture transport (Marshall, 2003).

A critical component in studying rainfall teleconnections is the quantification of moisture sources. (J. Li et al., 2024) demonstrate that even within a single region, heavy rainfall events can be fueled by vastly different moisture pathways. Using backward tracking models like WAM-2layers, researchers can identify whether

moisture is sourced from local land recycling or remote oceanic basins. In their study, Type 1 rainfall was dominated by moisture from the Western North Pacific (40.6 mm), while Type 2 relied more heavily on the Arabian Sea (18.5 mm) and Bay of Bengal (28.3 mm). For Malaysia, a similar approach could determine if changes in SIE favor specific moisture channels over others.

In their investigation of interhemispheric teleconnections,(Y. Li et al., 2019b) examine the role of stationary Rossby waves as primary drivers of atmospheric coupling between the Northern and Southern Hemispheres. While the equatorial region often serves as a "critical line" that reflects or absorbs wave energy, the authors utilize a wave ray ensemble (WRE) method to identify specific geographic "windows" and "barriers" that regulate this propagation. Their findings reveal that while the eastern Pacific and Atlantic act as consistent pathways for waves year-round, the Africa-Indian Ocean region only facilitates cross-equatorial transport during the boreal summer. Furthermore, the study demonstrates a "spectral-selective" filtering process, where the equator effectively blocks shorter waves and only permits the passage of longwaves with zonal wavenumbers less than six. This selective permeability underscores the complexity of how weather anomalies in one hemisphere can influence the climate patterns of the other.

Prabhu et al. (2021), examined the role of meridional circulation in linking Antarctic sea ice area (SIA) variability to tropical climate through latitude–pressure cross-sections of meridional wind and pressure vertical velocity (v , ω). Their results show that vertical velocity anomalies associated with Antarctic sea ice variability modulate tropical convection, demonstrating that meridional overturning circulation and vertical motion form a key atmospheric bridge through which high-latitude forcing influences tropical rainfall variability. Composite analyses of meridional and zonal

circulation further revealed that anomalously high Antarctic SIA is associated with large-scale circulation adjustments resembling the negative phase of the Southern Annular Mode, characterised by a subpolar high and a midlatitude low. These circulation anomalies generate coherent meridional overturning cells that extend from the Southern Hemisphere into the tropics.

This review (Stan et al., 2017) synthesizes the current understanding of interactions between the tropics and midlatitudes on intraseasonal time scales (10–100 days), a field increasingly recognized as a vital source of subseasonal to seasonal (S2S) predictability. Midlatitude dynamics play a significant role in triggering and modulating tropical weather. Upper-level extratropical disturbances, often described as potential vorticity (PV) "streamers" or tilted troughs, can penetrate the tropics through "westerly ducts" in the upper troposphere. These incursions are known to trigger deep convection in regions like the Intertropical Convergence Zone (ITCZ) and have been linked to the initiation of MJO events in the Indian Ocean. Furthermore, lower-tropospheric "cold surges" originating from midlatitude anticyclones (e.g., the Siberian High) can enhance air-sea interactions, promote tropical cyclogenesis, and inhibit rainfall event for short period by stabilizing the local environment with cold, dry air.

The study by (Hoskins & Karoly, 1981) shows that large-scale sources generate wavetrains in the upper troposphere. Long wavelengths (planetary-scale waves) have the capacity to propagate strongly poleward and eastward, often following "great circle" routes across the globe. Shorter wavelengths are more restricted; they tend to be trapped equatorward by the poleward flank of the jet stream, which causes a "split" in the wavetrains at middle latitudes. A key conclusion of the paper is that subtropical

forcing is the most efficient mechanism for producing a significant atmospheric response in middle and high latitudes.

Vertical motion in the atmosphere is governed primarily by large-scale patterns of horizontal divergence and convergence rather than by the magnitude of local wind speeds. According to quasi-geostrophic (QG) theory, vertical motion (often expressed as the change of speed in pressure coordinates height coordinates) is not simply a product of local wind speeds but is forced by specific large-scale dynamical imbalances (Holton & Hakim, 2013). Therefore, vertical motion in the atmosphere is not a simple consequence of the magnitude of local wind speeds, such as the zonal wind. Instead, it is forced by large-scale dynamical imbalances that arise as the atmosphere adjusts to maintain geostrophic and hydrostatic balance.

The paper (Azhar et al., 2023b) explains that when the amount of sea ice in the Antarctic's Indian Ocean sector changes, it sets off a "chain reaction" in the atmosphere that eventually reaches the tropics. This connection works like a series of waves, called Rossby waves, that travel through the air across the ocean. During a "high ice" year, there is more sea ice than usual create the cold conditions in the south help strengthen the Polar jet stream and large-scale wind patterns like the Polar, Ferrel, and Hadley cells.

Xiao et al., (2024) demonstrated that Antarctic sea ice anomalies influence East Asian precipitation primarily through atmospheric teleconnection mechanisms. Increased sea ice in the eastern Indian Ocean sector reduces upward turbulent heat flux, leading to lower-tropospheric cooling, enhanced atmospheric stability, and descending motion over Antarctica. This thermal forcing induces an Antarctic Oscillation–like circulation response, characterized by barotropic geopotential height and zonal wind anomalies that propagate toward East Asia via stationary Rossby waves and meridional

circulation adjustments. The downstream response includes coherent anomalies in zonal and meridional winds, vertical velocity, and vertically integrated moisture convergence/divergence, which regulate regional ascent and moisture supply and thereby modulate precipitation.

Makula & Zhou (2023), discussed the teleconnection process linking Antarctic sea ice concentration (SIC) to Tanzanian rainfall is driven by the propagation of atmospheric wave trains from high latitudes to the tropics. Positive SIC anomalies in the Weddell and Ross seas trigger upper-level wave activity that travels toward the southwestern Indian Ocean, establishing an anomalous cyclonic circulation in the upper troposphere. This upper-level feature induces low-level divergence and atmospheric subsidence over Tanzania, creating a dynamic environment that is highly unfavourable for precipitation. Besides, outgoing Longwave Radiation (OLR) serves as a critical indicator of the physical processes linking Antarctic sea ice to African climate, as it is used to monitor convective activity. In this context, positive OLR anomalies indicate a reduction in high-level cloud cover and signify suppressed convection or atmospheric subsidence. Conversely, negative anomalies would indicate enhanced convection and cloud development. The observed positive OLR patterns over Tanzania confirm that the sea ice-induced teleconnection physically inhibits the rising motion and moisture convergence necessary for cloud formation.

CHAPTER 3 METHODOLOGY

3.1 Study Area

Peninsular Malaysia and Antarctic Indian Ocean sector were selected to be investigated the potential Antarctic–tropical teleconnections, particularly the extent to which variability in Antarctic sea ice may be associated with changes in tropical precipitation over Peninsular Malaysia. Peninsular Malaysia is located in Southeast Asia between approximately 1° N–7° N and 99° E–105° E and experiences a humid tropical climate characterised by high temperatures, abundant moisture availability, and pronounced seasonal rainfall variability. Rainfall over Peninsular Malaysia is primarily influenced by large-scale atmospheric circulations, including the Northeast and Southwest Monsoons, the migration of the Intertropical Convergence Zone (ITCZ), and regional convection processes.

The Antarctic Indian Ocean sector forms part of the Southern Ocean surrounding Antarctica and is one of five major Antarctic sea-ice regions defined by the National Snow and Ice Data Center (NSIDC). This sector spans the longitudes roughly from 20° E to 90° E and exhibits strong seasonal and interannual variability in sea ice extent.

3.2 Raw data

The data used by this project are total precipitation, Antarctic sea ice extent, geopotential height, meridional wind, zonal wind, vertical velocity and vertical integrated moisture divergence. The study period is 35 years start from 1990 until 2024.

3.2.1 Precipitation data

Hourly total precipitation data were obtained from the ERA5 reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 provides globally consistent atmospheric and surface variables derived from a state-of-the-art data assimilation system that integrates model physics with a wide

range of observations. For ERA5 reanalysis data, precipitation is provided as an accumulated depth over the one-hour period ending at the valid time, with units expressed in metres of water equivalent. The data set has a spatial resolution of $0.25^\circ \times 0.25^\circ$.

3.2.2 Sea Ice Extent Data

Antarctic sea ice extent data were obtained from the National Snow and Ice Data Center (NSIDC), specifically from the dataset titled Sea Ice Index Regional Monthly Data (Ann Windnagel, 2020). This dataset provides monthly averaged sea ice extent and sea ice area from 1978 to the present, expressed in square kilometres, for five Antarctic regions which are Bellingshausen–Amundsen Sea, Indian Ocean, Pacific Ocean, Ross Sea and Weddell Sea. Monthly sea ice extent values provided by the NSIDC Sea Ice Index represent the arithmetic mean of daily sea ice extent estimates within each calendar month. Daily sea ice extent is derived from passive microwave satellite observations using a 15% sea ice concentration threshold, and monthly averaging is applied to reduce high-frequency variability and highlight seasonal to interannual changes.

3.2.3 Atmospheric Data

Monthly global atmospheric reanalysis data, sourced from the ECWMF fifth-generation reanalysis product (ERA5) spanning from 1979 to 2024. All the atmospheric data used is obtained from ERA 5 This data set has a spatial resolution of $0.25^\circ \times 0.25^\circ$.

Vertical velocity (ω) represents the speed of air motion in the vertical direction and is expressed in pressure coordinates (Pa s^{-1}). In the ECMWF Integrated Forecasting System, a pressure-based vertical coordinate system is employed, in which atmospheric pressure decreases with height. Upward air motion (ascent) is negative

value of vertical velocity while downward air motion(subsidence) is positive value of vertical velocity. The pressure levels used for this parameter are 1000, 975, 950, 925, 900, 875, 850, 825, 800, 775, 750, 700, 650, 600, 550, 500, 450, 400, 350, 300, 250, 225, 200, 175, 150, 125 and 100

The horizontal wind field in this study is represented by its zonal and meridional components, expressed in metres per second (m s^{-1}), following the standard meteorological convention used in the ECMWF Integrated Forecasting System. Zonal wind also known as u-component of wind, it is horizontal air movement in the direction of east or west. Positive u values indicate air flowing eastward, while negative u values indicate air flowing westward. Meridional wind is the v-component of wind that move northward (positive value) or southward (negative value). The pressure level used for the horizontal wind is same with the vertical velocity.

Vertically integrated moisture divergence represents the net horizontal spreading or accumulation of atmospheric moisture within an air column extending from the Earth's surface to the top of the atmosphere and is expressed in kg m^{-2} (equivalent to mm of water). Negative values indicate moisture convergence, where moisture accumulates within the column and favours upward motion, cloud formation, and precipitation, while positive values indicate moisture divergence, where moisture is removed and rainfall is suppressed.

Geopotential represents the gravitational potential energy per unit mass of air relative to mean sea level and is expressed in $\text{m}^2 \text{s}^{-2}$. In atmospheric analysis, geopotential at pressure levels is commonly interpreted through geopotential height, which is obtained by dividing geopotential by the Earth's gravitational acceleration ($g = 9.80665 \text{ m s}^{-2}$). In this study, geopotential fields at 200hPa, 500hPa, and 850 hPa are analysed to represent upper-, mid-, and lower-tropospheric circulation patterns,

respectively, and to diagnose large-scale pressure anomalies and wave-like structures associated with atmospheric dynamics.

3.3 Analysis Technique

Pearson correlation analysis and composite difference analysis are employed as the primary statistical methods in this study. The statistical significance of the Pearson correlation coefficients is assessed using the Student's t-test (Xiao et al., 2024), which is the standard approach for evaluating whether the correlation (or equivalently, the regression slope) differs significantly from zero.

Composite difference analysis is applied to atmospheric parameters to identify contrasts between high sea-ice years and low sea-ice years, following Prabhu et al. (2021). The statistical significance of composite differences is evaluated using a grid-point-wise two-sample t-test with unequal variances (Welch's t-test). This test is appropriate for climate composite analysis because it does not assume equal variances or equal sample sizes between the two groups (Daniel S. Wilks, 2006). At each grid point, the test independently evaluates whether the mean value of a given variable differs significantly between high and low sea-ice years.

Different t-tests are used because the analyses address distinct statistical questions. Welch's two-sample t-test is used for composite analysis to determine whether the mean states of two independent samples differ significantly. In contrast, Pearson correlation and linear regression analyses assess the strength and significance of linear relationships between variables, for which the Student's t-test provides an appropriate significance test for the regression slope.

Prior to statistical analysis, hourly total precipitation data are aggregated into monthly totals. These monthly precipitation values are then used to compute Pearson

correlation coefficients with Antarctic sea ice extent (SIE). Subsequent analyses are conducted for further discuss.

CHAPTER 4 RESULTS AND DISCUSSION

In this chapter, the relationship between Antarctic sea ice extent and the precipitation in peninsular Malaysia will be discussed. The impact of sea ice extent to the atmospheric circulation would be studied to discuss its mechanism for identify the precipitation pattern in tropic region.

4.1 Tropical Precipitation

Based on the review provided by (C. Zhang, 2005), tropical rainfall pattern is modulated by the features of MJO event which is connected to the overturning zonal circulation. During the active phase of MJO cause the air near to surface will converge and move eastward to the deep convective region known as active phase, while the suppressed phase occur at the east and west of the deep convergence centre, where sinking air leads to suppressed convection and drier conditions. Hence, the tropical precipitation for short term period is affected within few days to weeks if compared to long term period like ENSO effect is start from month to years.

The impact of sea ice on the precipitation variability can be identified by atmospheric parameter or oceanic parameter. Based on this paper (Hansen et al., 2010), the changing in sea surface temperature (SST) are smaller compared to the change in surface air temperature. This is because the ocean can store more heat compared to the atmospheric. The large-scale seasonal monsoon is affecting the rainfall variability but the intraseasonal oscillation and the atmospheric dynamic is another factor to control the rainfall event onset (H. Wang et al., 2018).

As mentioned by(Robertson et al., 2011), rainfall in the Maritime Continent is not merely a product of the seasonal monsoon but is heavily modulated by high-frequency atmospheric variables. The dominance of the diurnal cycle and the

occurrence of synoptic disturbances, such as cold surges and the Borneo Vortex, create significant rainfall variability that operates independently of long-term seasonal signals. Consequently, a zero-lag correlation is appropriate to capture the near-instantaneous response of precipitation to these rapidly evolving atmospheric drivers.

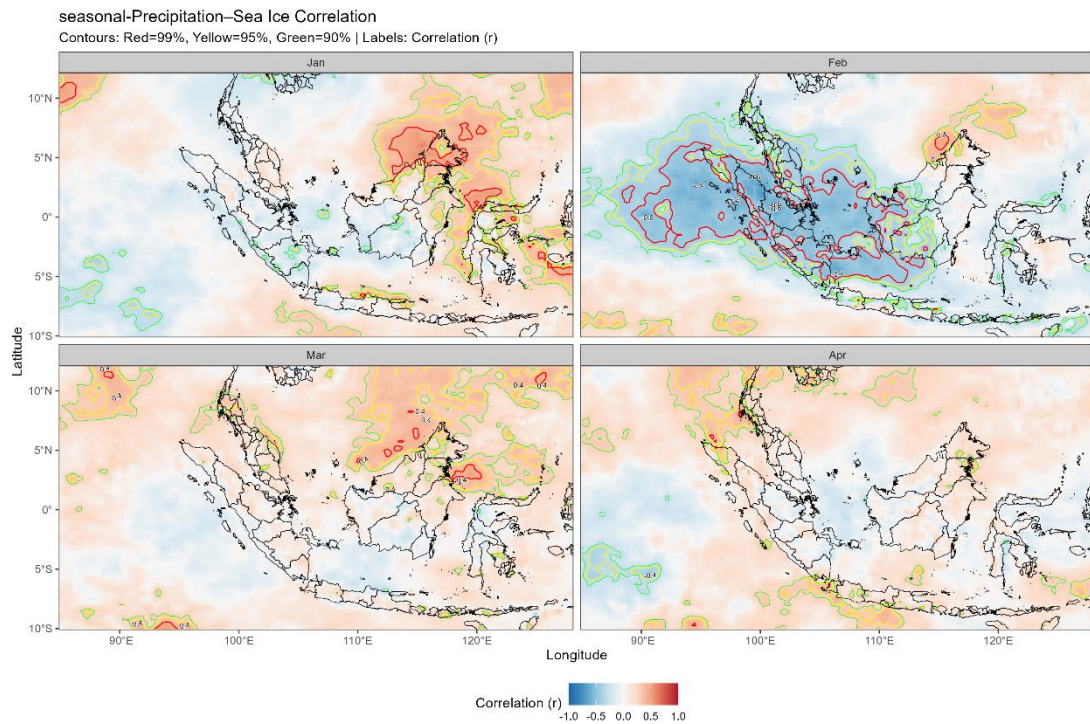
4.2 Correlation Between Precipitation and Antarctic SIE on February

Seasonal correlation maps (Figure 1) indicate that rainfall over Peninsular Malaysia exhibits statistically significant covariability with Antarctic sea-ice extent (SIE) during selected months. November, February, July, and August show correlations exceeding the 95% confidence level. Among these, February stands out as the month with the most spatially coherent and statistically robust signal, characterized predominantly by negative correlations across Peninsular Malaysia, indicating that higher Antarctic SIE is associated with reduced rainfall. The spatial pattern of the correlation evolves seasonally. In November, negative correlations are concentrated over northern Peninsular Malaysia, while positive correlations dominate the southern region. By January, the pattern further differentiates, with positive correlations emerging along the east coast and negative correlations along the west coast of the Peninsula. This rapid spatial reorganization of the rainfall–SIE relationship suggests that the Malaysian atmosphere responds sensitively to large-scale circulation adjustments, consistent with a dynamically driven teleconnection rather than a delayed oceanic response.

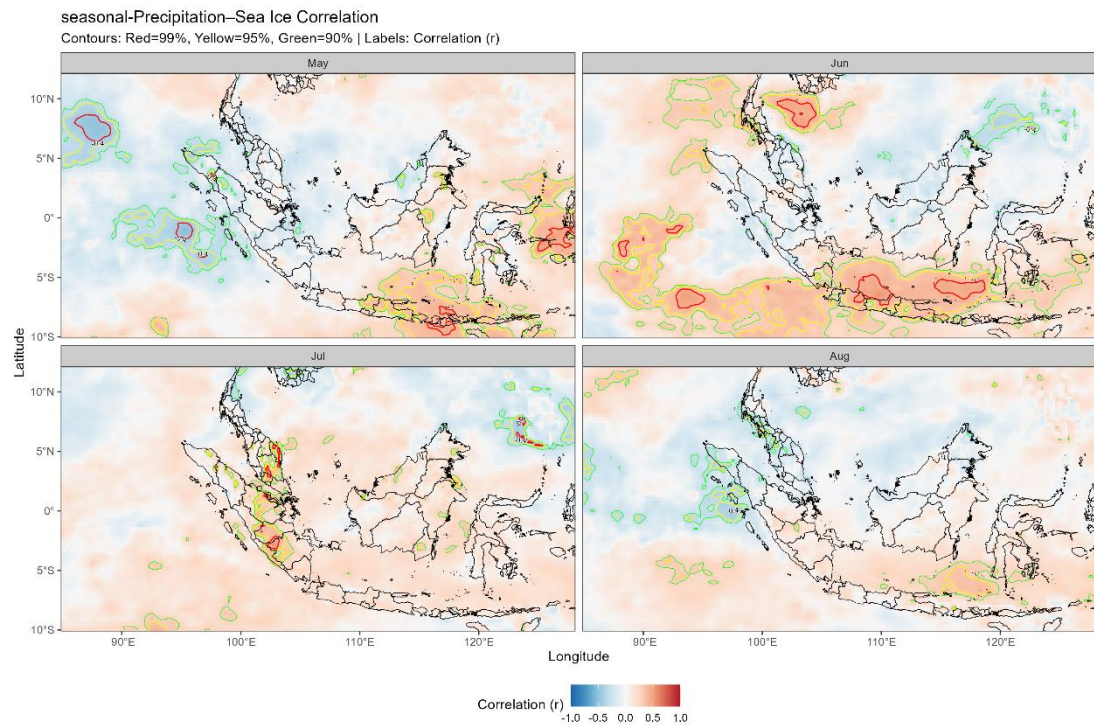
The enhanced February signal reflects the late boreal winter atmospheric background state, which large-scale circulation anomalies become increasingly organized and dynamically coherent. Atmospheric responses to high-latitude forcing tend to strengthen progressively from early to late winter, reaching peak detectability in February (W. Zhang & Jiang, 2023). Consistently, higher circulation prediction skill

in February compared to January has been reported by (Qiao et al., 2020), highlighting the greater stability and persistence of planetary-scale flow patterns during this period. These findings support the physical plausibility of the stronger February signals identified in this study.

a)



b)



c)

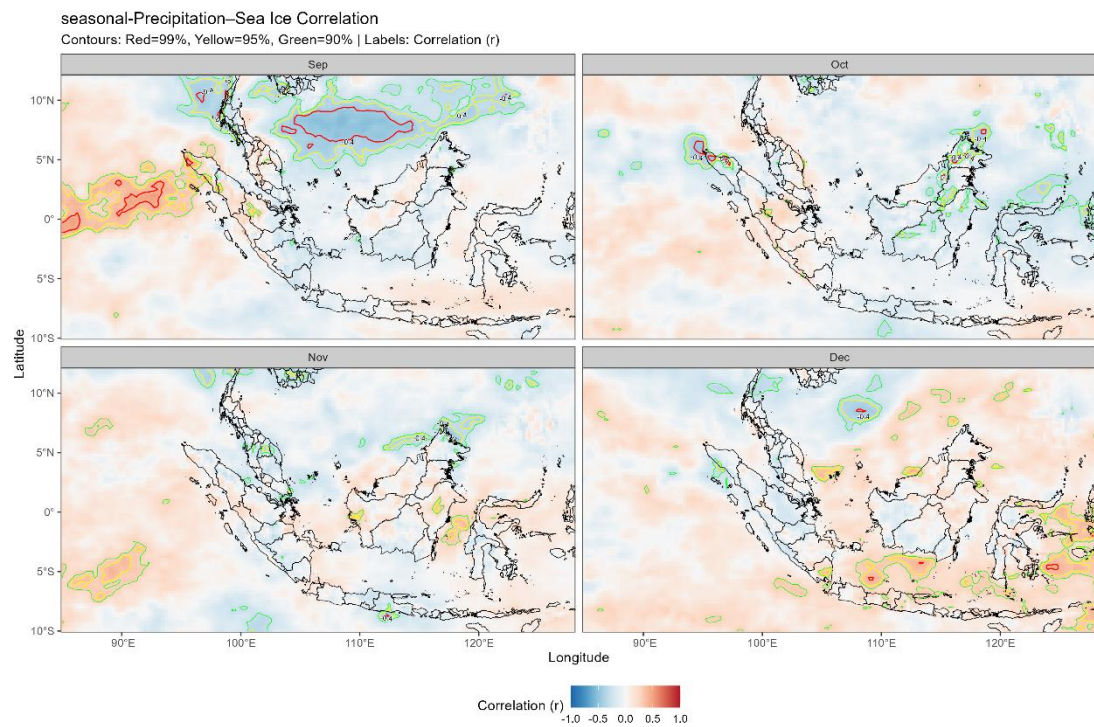


Figure 1 Correlation between the Antarctic SIE and the Precipitation in Malaysia, same for a, b and c. a) January until April, b) May until August and d) September until December. Green, Yellow and red contour line indicate the values that significantly exceed the 90%, 95% and 99% confidence level, respectively

4.2.1 Dynamic Mechanism in February

Antarctic sea-ice extent (SIE) anomalies alter the atmospheric and ocean parameter by increasing surface albedo and reducing ocean–atmosphere turbulent heat fluxes. (Hines et al., 2015; Wu & zhang, 2011). Hence, the positive anomalies in Antarctic SIE can affect the surface heat flux and meridional temperature gradient over the Indian Ocean Sector via the reduction of ocean-atmospheric heat flux and increase the albedo effect (Y. Zhang & Semtner, 2001). Therefore, the polar atmosphere is cool and the air become denser cause the surface pressure increases at high latitudes. The contrast of high polar pressure and low mid-latitude pressure strengthen the meridional pressure gradient. The stronger the meridional pressure gradient, the stronger the zonal (westerlies) wind produced (Marshall, 2003). This situation shows the primary dynamical pathway through which Antarctic sea ice anomalies influence the large-scale atmospheric circulation. Antarctic sea ice extent may be indirectly influenced rainfall in Peninsular Malaysia through atmospheric pathway. Barotropic structure is influenced by the anomalous cyclonic and anticyclonic also create the anomalous geopotential height.(Chandra & Sandeep, 2024)

The intensification of the westerlies provides a favourable waveguide for Rossby wave activity. Evidence for a Rossby-wave–like response is clearly seen in Figure 2, where February geopotential height anomalies at 200 hPa, 500 hPa, and 850 hPa exhibit a vertically coherent, barotropic structure. The atmospheric response has an equivalent barotropic structure, meaning the wave patterns have the same sign throughout the depth of the atmosphere (Hoskins & Karoly, 1981). Negative geopotential height anomalies are observed over the polar region, positive anomalies dominate the mid-latitudes, and negative anomalies re-emerge toward the low latitudes, forming an alternating ridge–trough pattern characteristic of large-scale Rossby waves. These anomalies are statistically significant at the 95% confidence level over the Indian Ocean sector. The use of the composite difference analysis, together with the absence of longitudinal phase propagation, indicates the presence of a stationary Rossby wave rather than a propagating Rossby wave (Xiao et al., 2024).

Although geopotential height anomalies over Peninsular Malaysia are weakly negative, the regional circulation remains anticyclonic because the area is located on the equatorward flank of a larger midlatitude ridge–trough system. In such large-scale wave patterns, the direction of upper-level winds is controlled by horizontal gradients in geopotential height rather than by the sign of local height anomalies alone. These gradients produce clockwise (anticyclonic) flow over Peninsular Malaysia. As the region lies within the subsident sector of the upper-level circulation associated with the ridge–trough system, upper-level convergence occurs, which induces downward motion through mass continuity. This subsidence suppresses deep convection and rainfall, although the presence of an upper-level trough elsewhere in the circulation pattern. Air flows along geopotential height gradients and curves due to Earth’s rotation, resulting in clockwise or anticlockwise circulation depending on the large-scale pressure field.

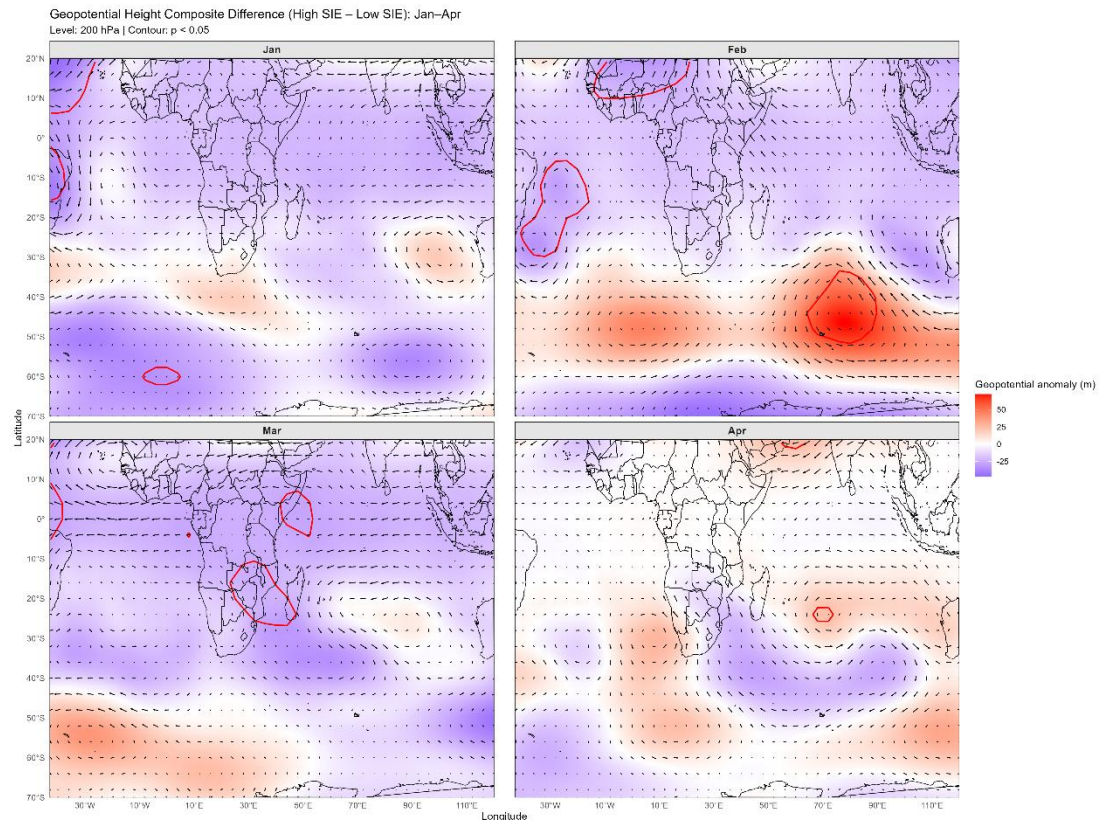
In mid-latitude atmospheric dynamics, regions ahead of an upper-level trough are typically associated with rising motion, while regions behind the trough favour subsidence. This occurs because air tends to diverge at upper levels downstream of the trough, inducing upward motion below, whereas convergence aloft upstream forces air to sink in order to conserve mass (Holton & Hakim, 2013; John M. Wallace & Peter V. Hobbs, 2006)

When air converges at upper levels, it cannot accumulate and is therefore forced to sink, leading to subsidence over Peninsular Malaysia despite the presence of weakly negative geopotential height anomalies. This subsiding motion extends through much of the tropospheric column. Vertical velocity and meridional wind composites (Figure 3) support this interpretation, showing statistically significant positive ω anomalies over Peninsular Malaysia in February from 1000 hPa up to 200

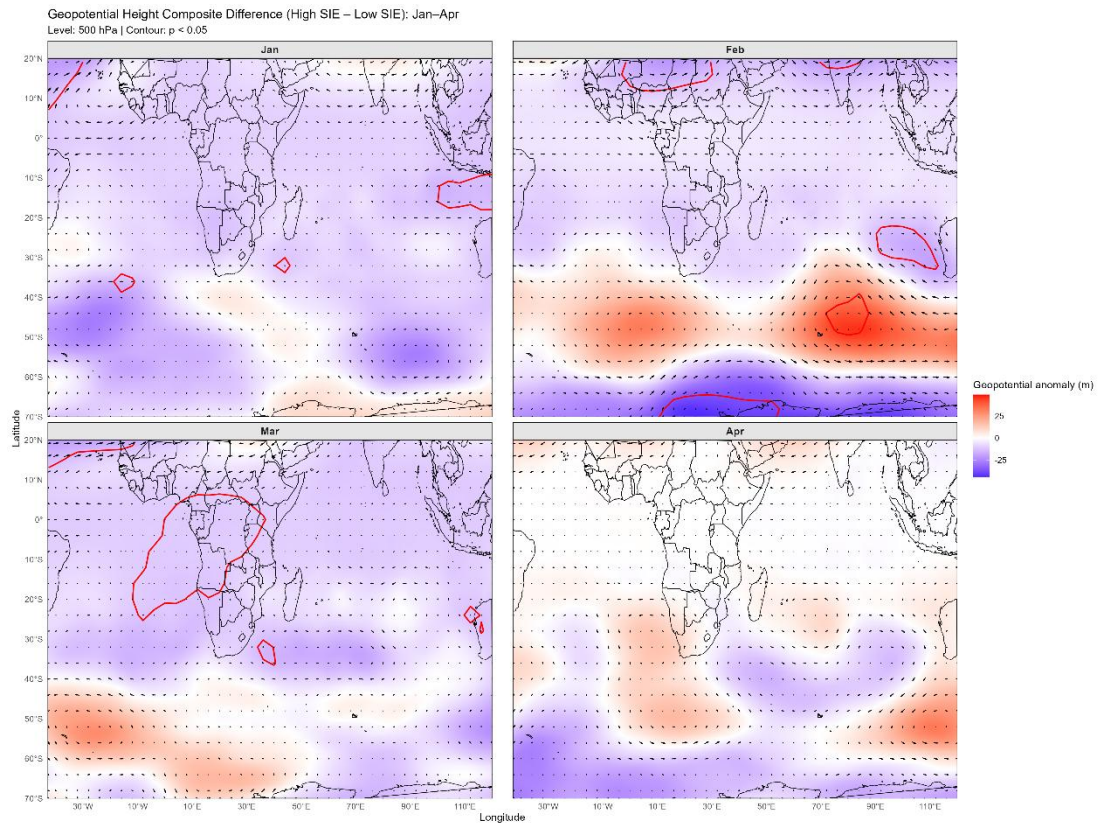
hPa. These positive ω anomalies indicate downward motion of air, which suppresses ascent, limits cloud development, and reduces convective rainfall rather than enhancing it.

The low potential height is show and promote the horizontal circulation and the air rise occur in low pressure and vice versa for sink air. Importantly, vertical motion in the atmosphere is not determined solely by pressure or geopotential height anomalies but arises dynamically from the convergence and divergence of the horizontal wind field (John M. Wallace & Peter V. Hobbs, 2006). Additionally, the vertical integrated moisture divergence in Figure 5 shows positive anomalies which enhance the fact of subsidence in Peninsular Malaysia due to the moisture is exported out of the Peninsular Malaysia and the air column tends to dry which reduce the rainfall event. The vertical motion and omega are the key factors to identify the rainfall pattern either subsidence or sinking.

a)



b)



c)

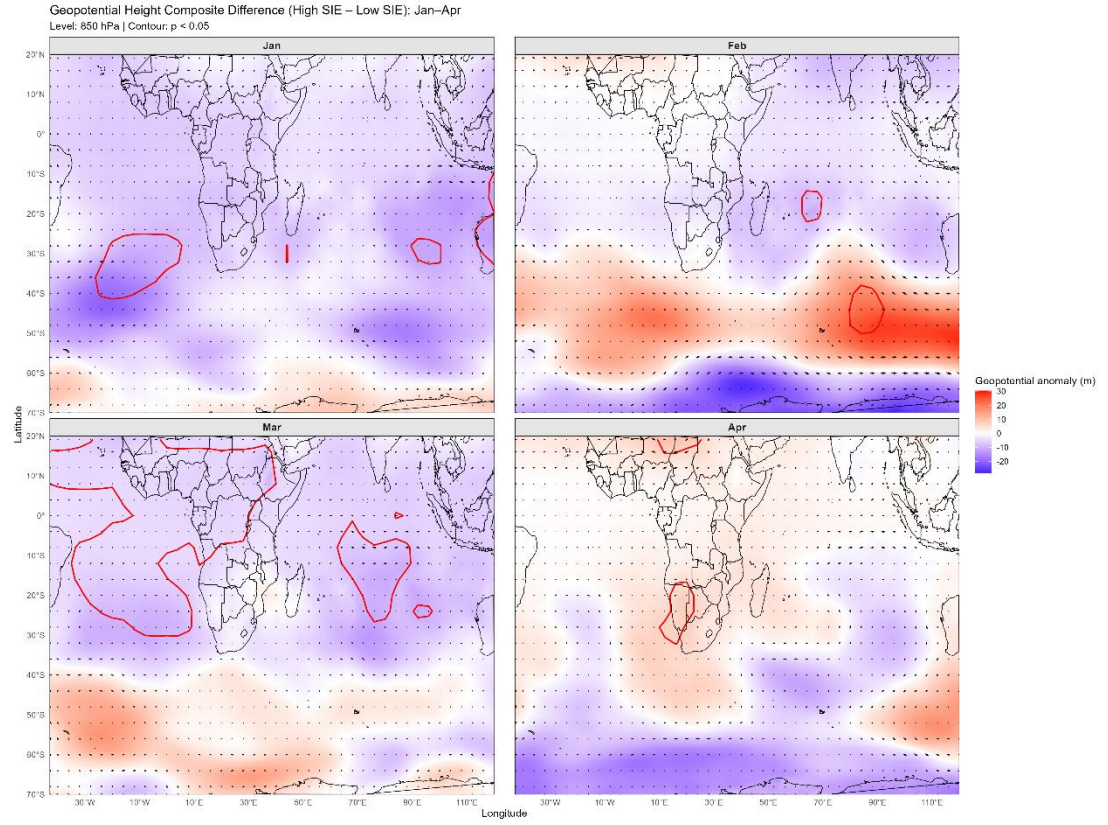
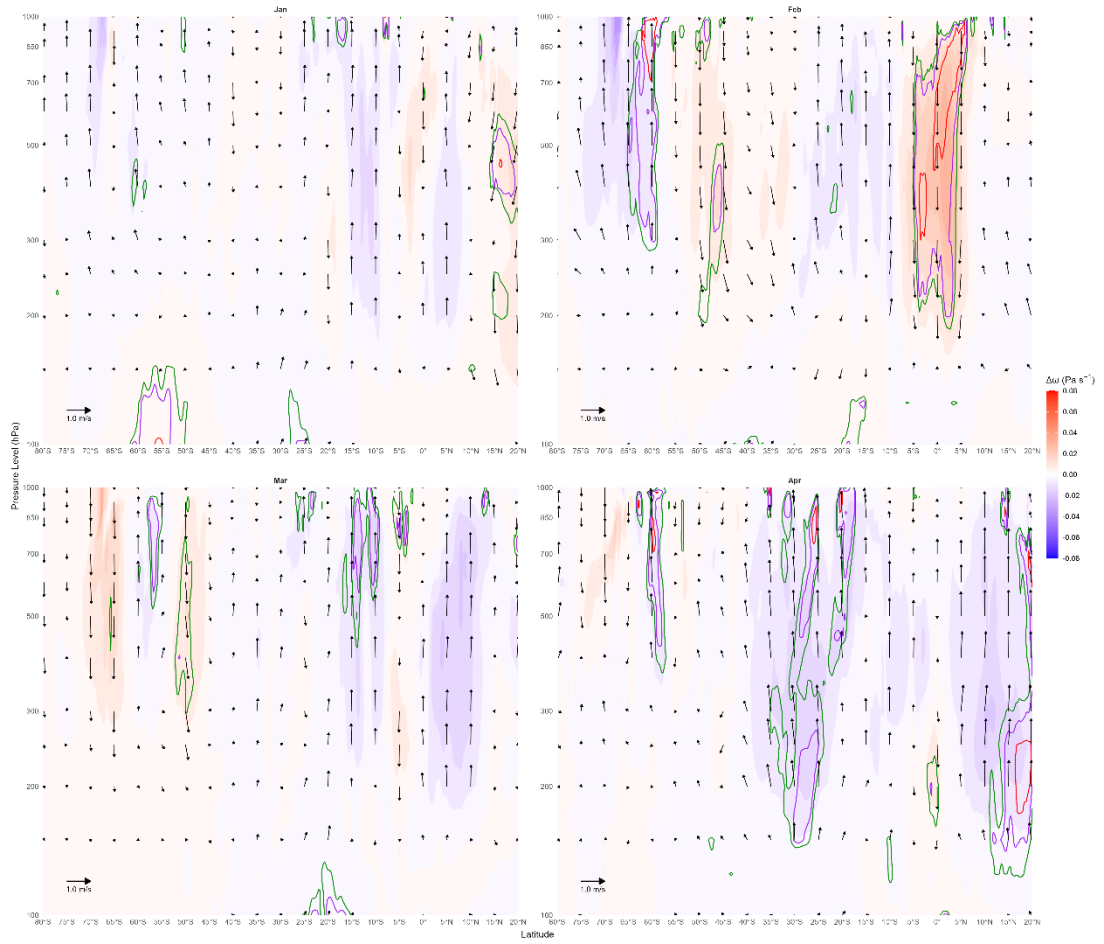


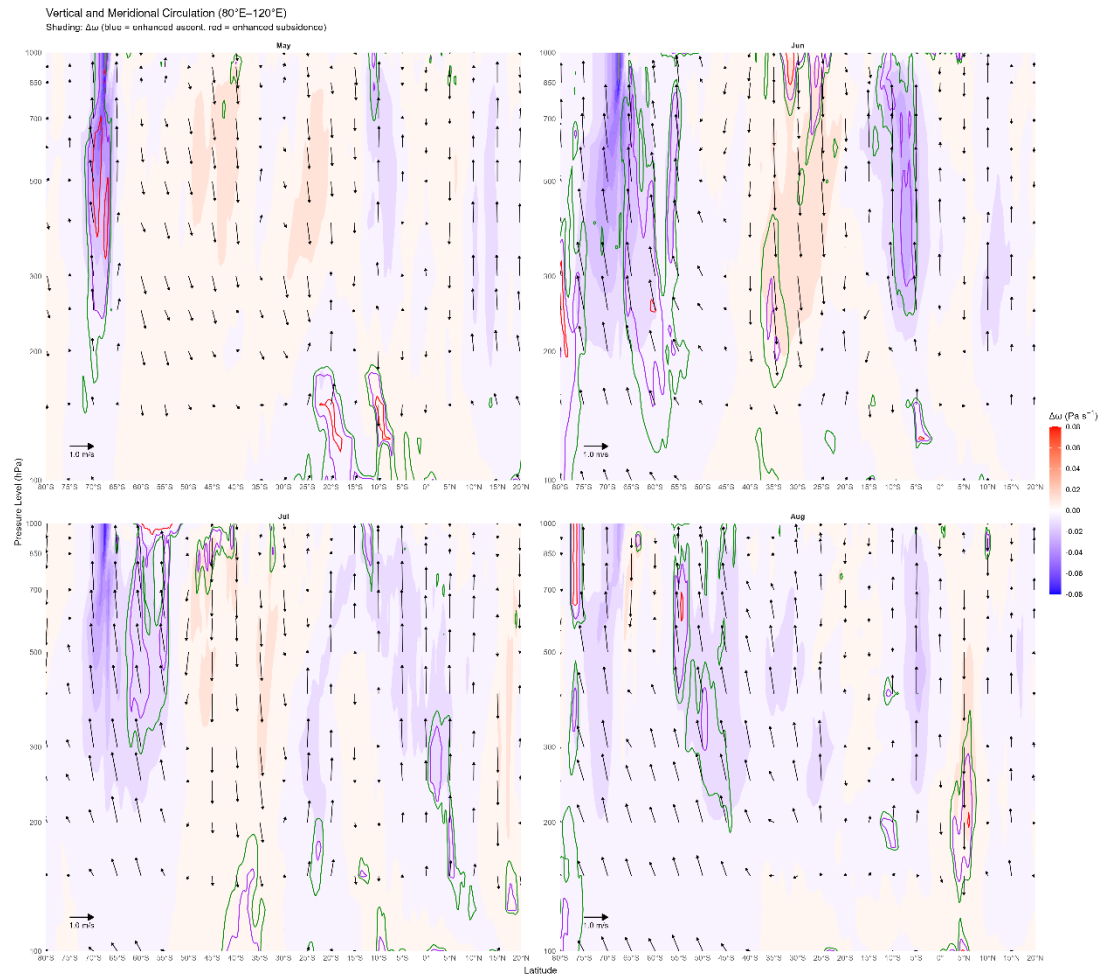
Figure 2 Geopotential height of January until April (a) 200 hPa, (b) 500 hPa, (c) 850 hPa geopotential height (unit: m; shaded) and horizontal wind (unit: $m s^{-1}$; vectors) during 1990–2024. The red contour indicates the values that significantly exceed the 95% and 90% confidence level, respectively.

a)

Vertical and Meridional Circulation (80°E–120°E)
Shading: Δw (blue = enhanced ascent, red = enhanced subsidence)



b)



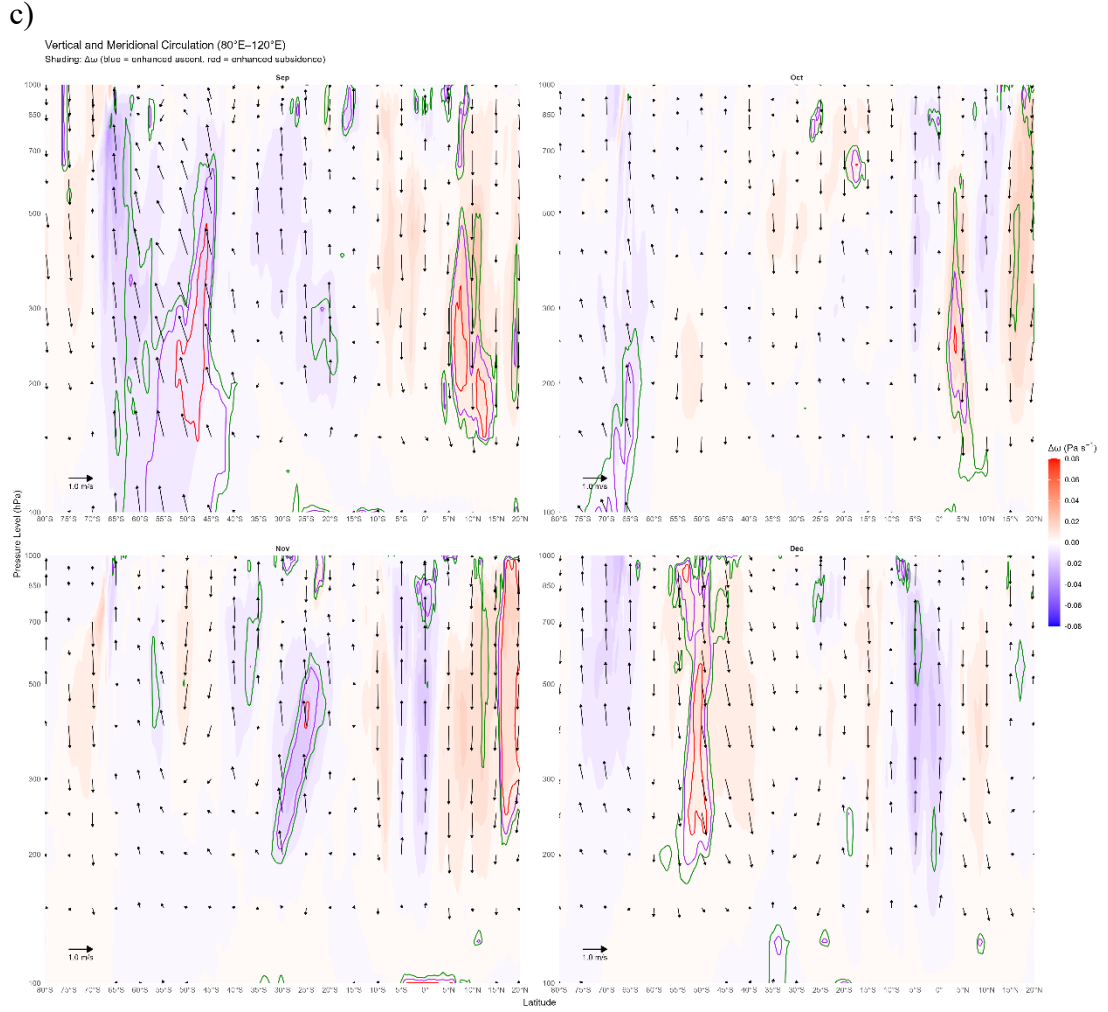


Figure 3 Vertical-horizontal cross section for vertical wind (unit: $m s^{-1}$; vector) and omega (unit: $-10^{-2} Pa s^{-1}$; shaded) (a) Jan-Apr; (b) May-Aug; (c) Sep – Dec separately. Green, Yellow and red contour line indicate the values that significantly exceed the 90%, 95% and 99% confidence level, respectively

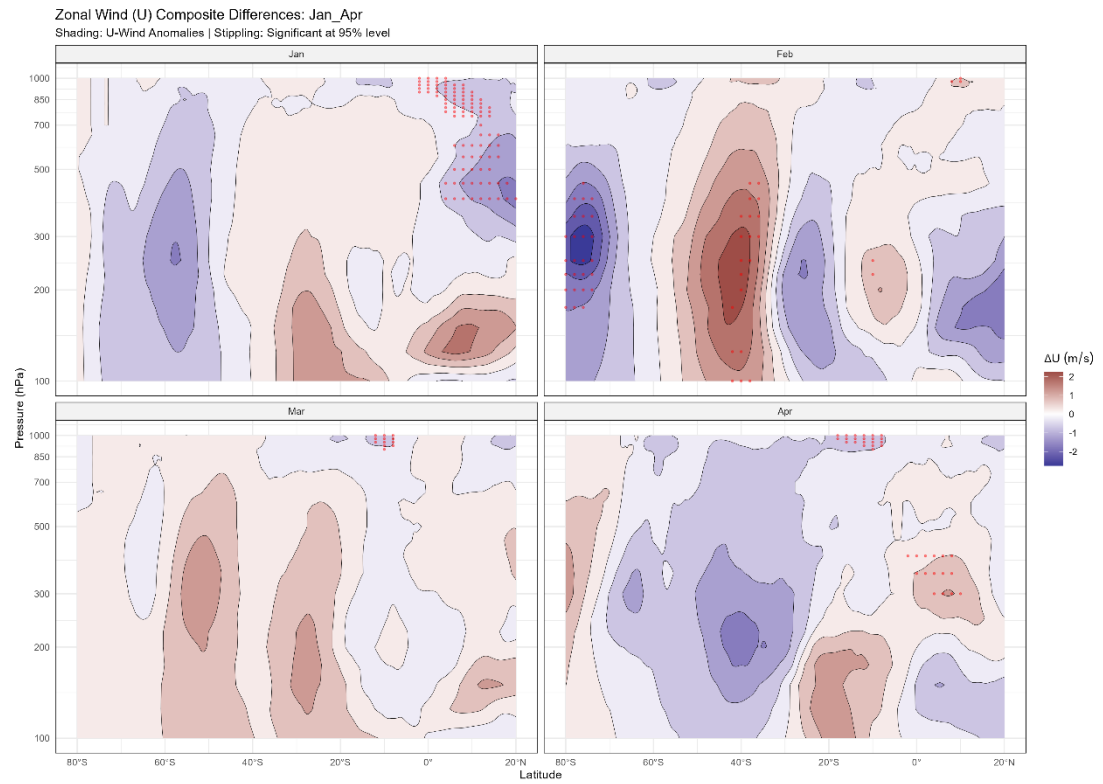
Zonal wind composites (Figure 4) further support the subsidence interpretation by revealing a clear vertical contrast in the zonal wind response. Positive zonal wind anomalies in the middle and lower troposphere indicate anomalous westerly flow, whereas negative anomalies in the upper troposphere reflect a weakening of westerlies aloft. However, zonal wind anomalies over Peninsular Malaysia are relatively weak, indicating that the regional circulation response is not driven by local jet intensification. However, the strong mid-latitude westerlies indirectly regulate the large-scale circulation by modifying geopotential height anomalies, which then alter the tropical vertical motion. Although zonal wind anomalies over the northern tropics are weak, tropical vertical motion can still be strongly modulated by remote midlatitude forcing.

Strengthened midlatitude westerlies modify geopotential height and Rossby wave patterns, producing upper-level convergence and subsidence over the tropics without requiring local jet intensification (Holton & Hakim, 2013; John M. Wallace & Peter V. Hobbs, 2006). This highlights the indirect nature of the Antarctic–tropical teleconnection, whereby changes in the midlatitude jet modulate tropical circulation and rainfall through large-scale atmospheric adjustments.

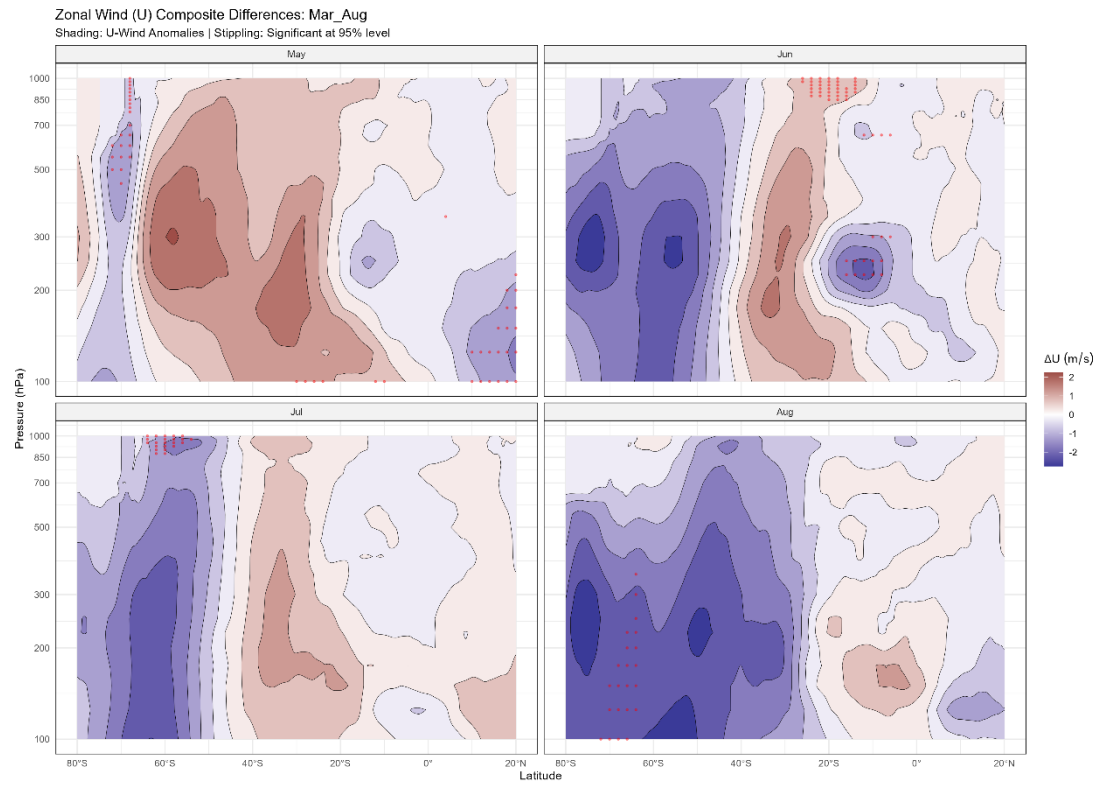
Subsidence plays a dominant role in regulating tropical rainfall by suppressing deep convection and moisture convergence. As demonstrated by Xiao et al. (2024), Antarctic sea-ice–induced circulation anomalies influence precipitation primarily through modifications to vertical motion and column moisture convergence, rather than through direct surface forcing. In this study, the diagnosed subsidence over Peninsular Malaysia weakens low-level moisture convergence, thereby limiting the moisture supply required for sustained convective development. Although the circulation anomalies are associated with an upper-level trough, Peninsular Malaysia is situated downstream of this feature, where quasi-geostrophic theory predicts upper-level convergence rather than divergence, leading to subsidence through mass continuity. Moreover, rainfall over Peninsular Malaysia during the mature Northeast Monsoon is particularly sensitive to large-scale subsidence, thereby enhancing the detectability of this Antarctic–tropical atmospheric teleconnection.

Overall, the February rainfall–SIE relationship is dynamically consistent with Antarctic sea-ice–induced circulation anomalies that promote upper-level convergence, tropospheric subsidence, and reduced moisture convergence over Peninsular Malaysia. This confirms that the negative correlation arises from large-scale atmospheric dynamics instead of direct polar forcing and highlights the strong seasonal dependence of Antarctic and tropical teleconnections.

(a)



(b)



(c)

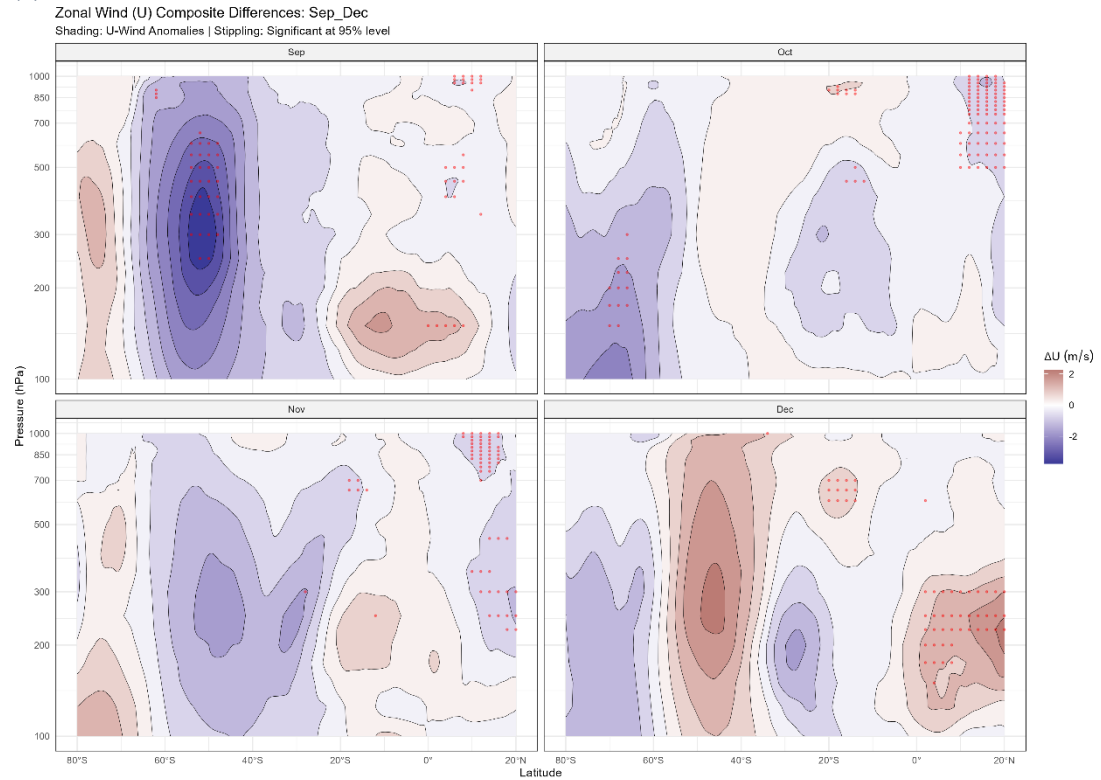
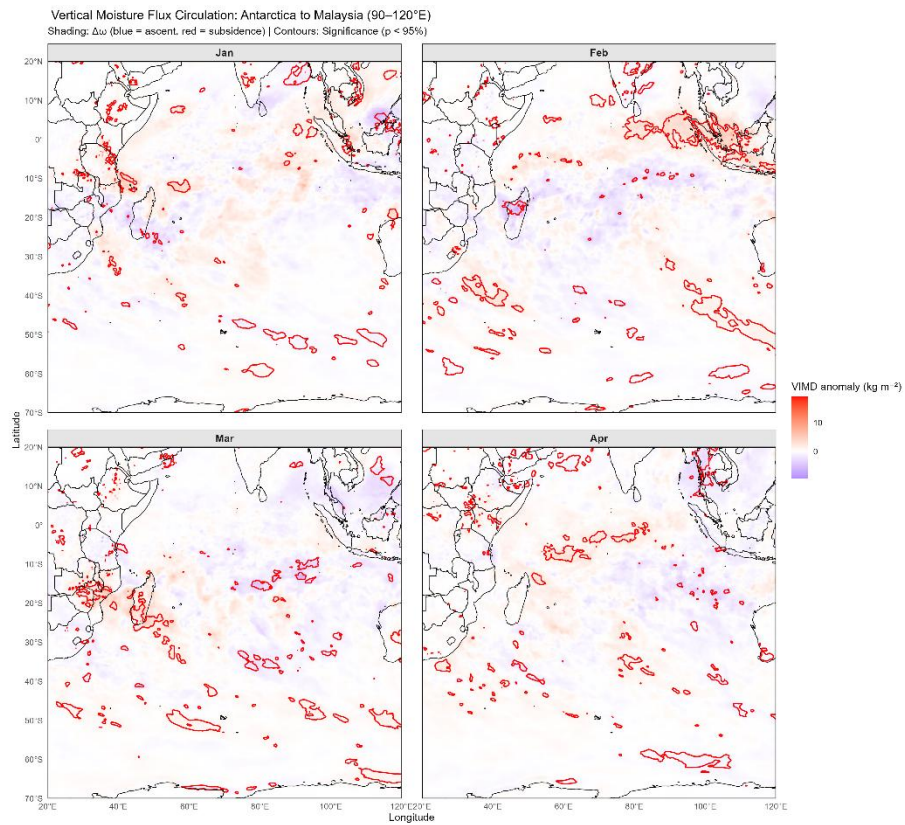


Figure 4 Zonal wind (unit: $m s^{-1}$; shaded) anomalies along $80^{\circ}S-120^{\circ}E$ for (a) Jan-Apr; (b) May-Aug; (c) Sep-Dec within period 1990-2024. Red stippling indicates the values that significantly exceed the 95% confidence level

(a)



(b)

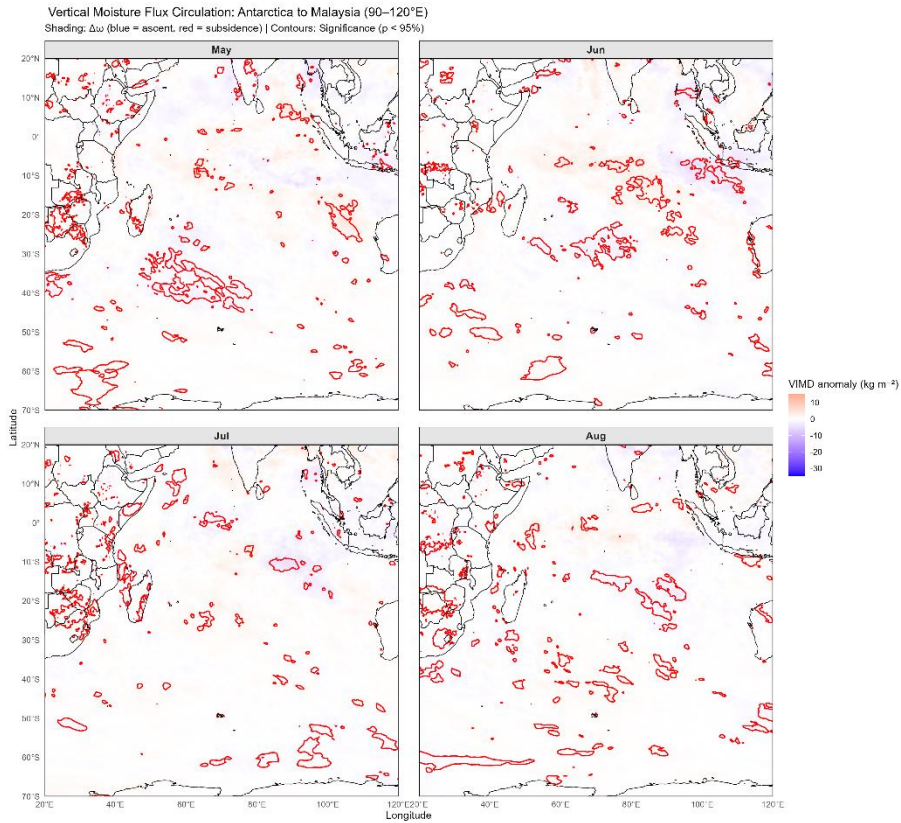


Figure 5 Vertical integrated moisture flux (kg m^{-2}) for (a) Jan-Apr; (b) May-Aug within period 1990–2024. Red contour indicates the values that significantly exceed the 95% confidence level

4.3 Correlation Between Precipitation and Antarctic SIE on July and August

The correlation analysis indicates that the relationship between Antarctic sea-ice extent (SIE) and precipitation over Peninsular Malaysia is stronger and more spatially coherent in July than in August. In July, a positive correlation is observed across much of Peninsular Malaysia, suggesting that Antarctic sea-ice variability may weakly modulate regional rainfall during this period because February show more stronger correlation pattern (Figure1). In contrast, August shows a more heterogeneous pattern, with significant negative correlations confined mainly to the northern part of Peninsular Malaysia and no significant positive correlation in the southern region. This contrast suggests a progressive weakening of the large-scale teleconnection from July to August, consistent with increasing dominance of regional monsoon processes later in the season.

4.3.1 Dynamic Mechanism in July

July corresponds to the mature phase of the Southwest Monsoon, so intense solar heating over the Asian landmass establishes a strong hemispheric thermal contrast (Yihui & Chan, 2005). This contrast drives cross-equatorial flow from the Southern Hemisphere, which turns into low-level southwesterlies after crossing the equator, transporting substantial moisture from the Indian Ocean toward Peninsular Malaysia. The northward displacement of the Intertropical Convergence Zone (ITCZ) further enhances convective activity, forming the climatological background state for peak monsoonal rainfall. The hemispheric thermal contrast will contribute the either side of rainfall especially for boreal winter rainfall and boreal summer rainfall (Chang et al., 2005). The author also mentioned the rainfall event shift northward during July is due to the shift ITCZ (Schneider et al., 2014, Chang et al., 2005). It also supports the July result in Figure 4 that wind starts as an easterly flow in the south (blue shading), but the 'turning' process at the equator transforms it into the low-level southwesterlies (red shading) by the time it reaches Malaysia.

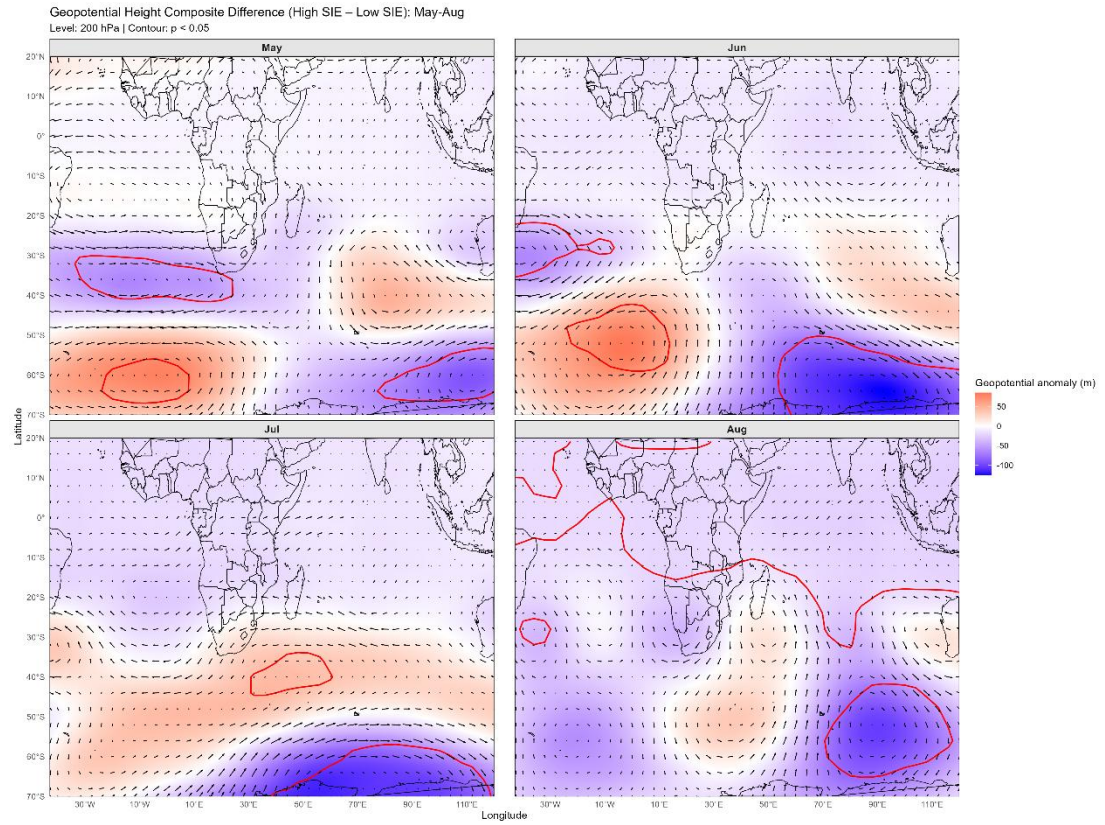
Under this strong background circulation, Antarctic SIE does not act as a primary driver of rainfall variability. Instead, SIE anomalies provide a secondary, modulatory influence. Higher Antarctic SIE increases surface albedo over the Southern Ocean, slightly reducing ocean–atmosphere heat fluxes and maintaining cooler high-latitude conditions in the Southern Hemisphere. This process marginally enhances the hemispheric thermal gradient, which can weakly reinforce cross-equatorial energy transport and the northward tendency of the ITCZ.

However, the dynamical fields indicate that this modulation remains limited. The geopotential height composites at 200 hPa (Figure 6) reveal a weak and poorly defined Rossby wave pattern during July compared to February (Y. Li et al., 2019b).

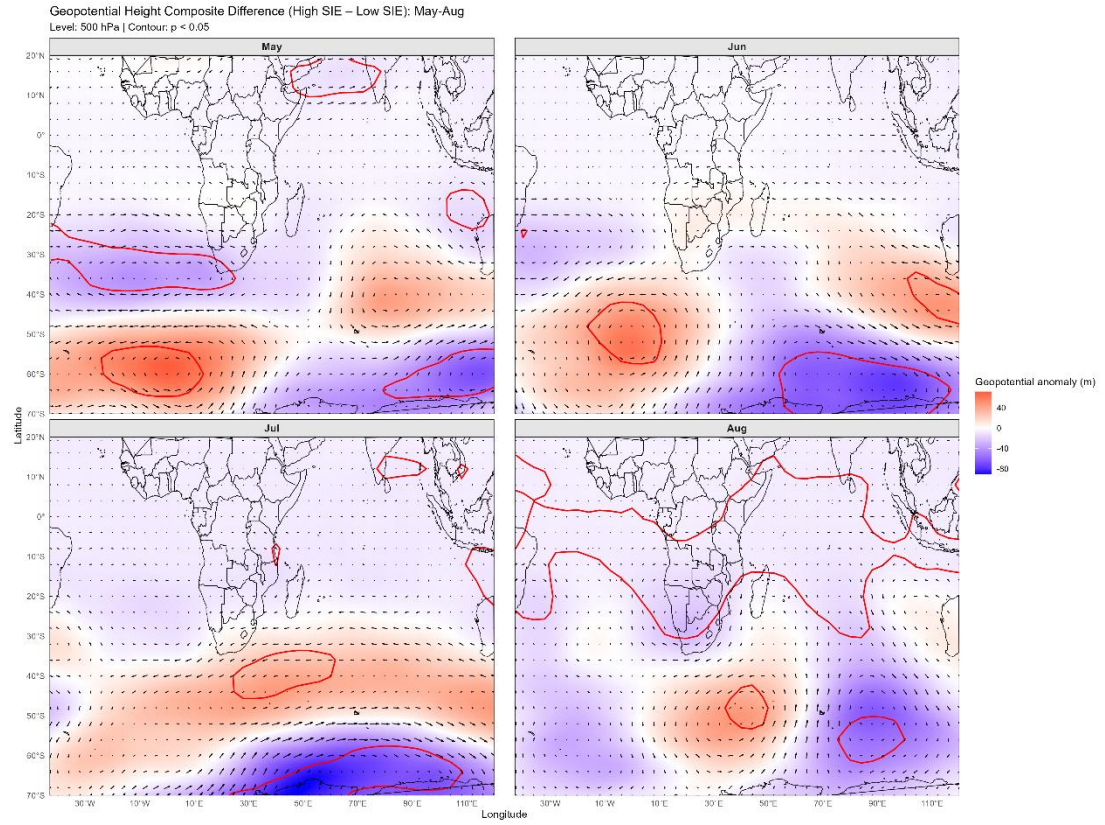
the author mentioned the value of waves moving from the Northern Hemisphere to the Southern Hemisphere CEWs (N_{N28}) is high during June–October and low during boreal winter months⁵. Although alternating geopotential anomalies are present in the Southern Hemisphere midlatitudes, their amplitude is reduced, and the wave energy decays rapidly toward the tropics. This behaviour is consistent with the seasonal northward shift of the jet stream and reduced waveguiding efficiency during boreal summer, which limits equatorward propagation of extratropical disturbances.

The vertical velocity and meridional circulation (Figure 3) further supports this interpretation. Negative vertical velocity anomalies (positive ω , subsidence) appear intermittently in the middle and lower troposphere during high-SIE years, indicating a tendency toward suppressed ascent. Nevertheless, this subsidence signal is weak and spatially discontinuous, suggesting that it is partially overridden by strong monsoonal convection and moisture convergence. As a result, rainfall variability in July reflects a balance between dominant regional monsoon forcing and weak extratropical modulation, producing the observed positive but modest SIE–precipitation correlation.

(a)



(b)



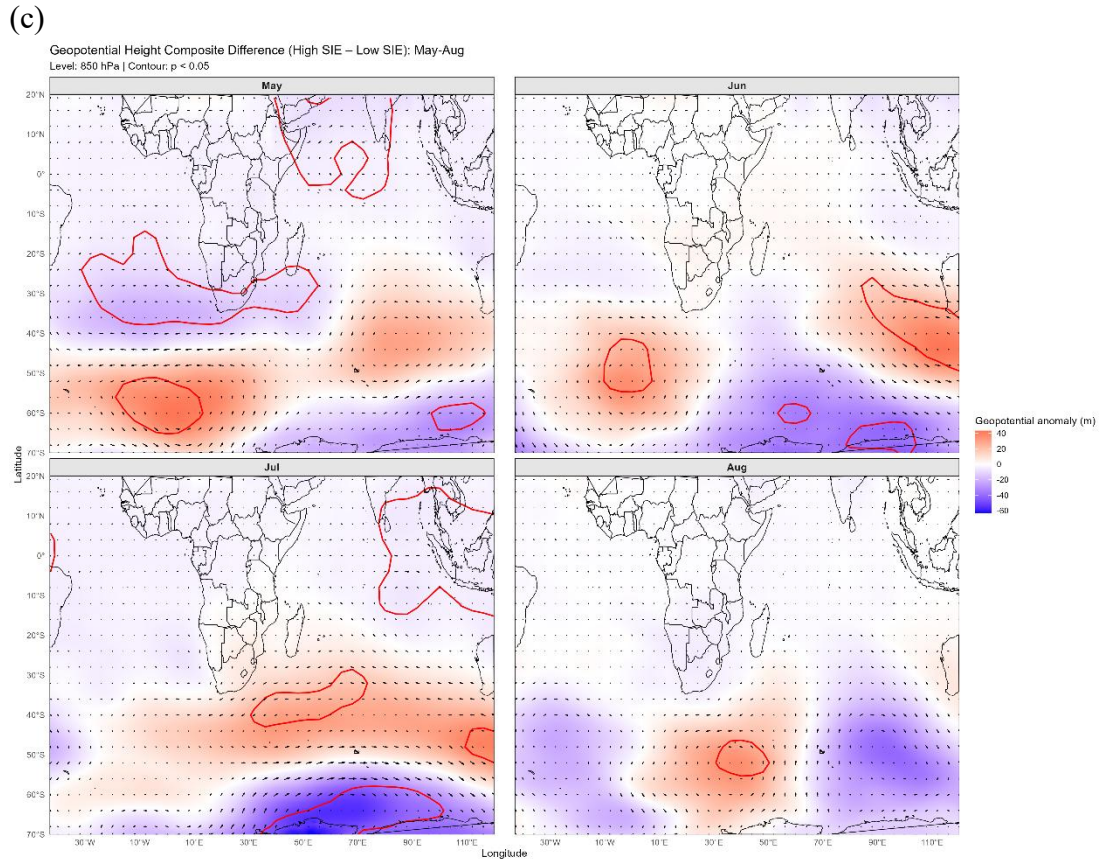


Figure 6 Geopotential height at (a) 200hPa, (b)500hPa and (c) 850hPa for May until August. Red contour indicates the values that significantly exceed the 95confidence level

Overall, the July–August analysis indicates that Antarctic sea-ice extent exerts only a weak and seasonally constrained influence on precipitation over Peninsular Malaysia. In July, a modest positive correlation emerges under the mature Southwest Monsoon, reflecting a limited modulatory effect of Antarctic SIE acting through hemispheric thermal contrast and weak upper-tropospheric dynamical adjustments. However, this influence remains secondary to strong monsoonal moisture transport and convection, which dominate the regional rainfall response. By August, the correlation weakens and becomes spatially heterogeneous, indicating a further reduction in extratropical influence as local monsoon processes intensify. These findings demonstrate that Antarctic SIE–rainfall linkages over Peninsular Malaysia are regime-dependent, with teleconnection signals becoming progressively suppressed during peak monsoon conditions.

CHAPTER 5 CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusion

Rainfall is an effective “detective” for large-scale circulation anomalies. This study demonstrates that Antarctic sea ice extent (SIE) in the Indian Ocean sector is significantly linked to rainfall variability over Peninsular Malaysia through large-scale atmospheric teleconnections. The relationship is strongly significant during February that exhibiting the most coherent and statistically robust signal. During this late boreal winter period, the signal is detectability due to the transition of large-scale circulation. Based on the correlation result shows that higher Antarctic SIE is consistently associated with reduced rainfall over much of Peninsular Malaysia, indicating that polar climate variability can influence tropical precipitation despite the large geographical separation.

Dynamical analyses confirm that this linkage is governed primarily by atmospheric circulation adjustments rather than local thermodynamic effects. Enhanced Antarctic SIE strengthens meridional temperature and pressure gradients, reinforcing midlatitude westerlies and enabling stationary Rossby wave propagation toward lower latitudes. These wave-induced circulation anomalies produce upper-level convergence, tropospheric subsidence, and moisture divergence over Peninsular Malaysia, thereby suppressing deep convection and rainfall. The results highlight that vertical motion and column moisture convergence are the key mechanisms through which Antarctic SIE modulates tropical rainfall.

In contrast, the Antarctic SIE–rainfall relationship during July and August is weaker and less spatially coherent, reflecting the dominance of regional monsoon dynamics during boreal summer. Under strong Southwest Monsoon forcing, extratropical signals act only as secondary modulators and are increasingly masked by

intense moisture transport and convection. Overall, this study shows that the influence of Antarctic sea ice on rainfall in Peninsular Malaysia does not occur all the time, but mainly during certain seasons when the tropical atmosphere is more responsive to large-scale circulation changes. This highlights that changes in Antarctic sea ice should be considered when trying to understand and predict seasonal rainfall variability in Peninsular Malaysia.

5.2 Recommendation

Future studies should include more oceanic parameters, such as sea surface temperature and ocean–atmosphere heat fluxes, to better understand how Antarctic sea ice interacts with the climate system. Analysing these variables may help explain why the relationship between Antarctic sea ice and rainfall in Peninsular Malaysia is weak or not significant in some months.

In addition, future research should examine Rossby wave characteristics, such as wave number and propagation behaviour, and consider other major atmospheric drivers, especially ENSO. Accounting for these mechanisms would help isolate the specific role of Antarctic sea ice and improve the interpretation of tropical rainfall variability.

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